

YESSICA SULE

Mumbai, India

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Education

Sardar Patel Institute of Technology

B.Tech in Computer Science Engineering

Aug 2023 – July 2027

Mumbai, Maharashtra

Prakash College of Commerce and Science

MHT-CET percentile: 99.52%ile

July 2021 – June 2023

Mumbai, Maharashtra

Relevant Coursework

- Data Structures & Algorithms
- Database Management
- Computer Networks
- Artificial Intelligence
- Machine Learning
- Computer Vision
- Internet of Things

Experience

Wired Lab, Indian Institute of Technology Bombay (IITB)

Feb. 2026 – July 2026

Software Engineer Research Intern

Mumbai, Maharashtra

- Designed a real-time **Unity 3D Digital Twin** driven by custom C# UDP socket listeners, animating a 4-avatar side-by-side comparative playback scene at 30+ FPS.
- Built a modular Python-to-Unity socket communication protocol streaming synchronized multi-model joint angles, confidence scores, and real-time uncertainty metrics.
- Created an interactive calibration wizard and real-time CSV data logging/visualization suite using SciPy and Matplotlib for dynamic time-series kinematic analysis.

Bharat Space Education Research Centre (BSERC)

June 2026 – July 2026

Defence-Space Summer Intern

Remote

- Studied **Generative AI, Cyber Security & Digital Forensics**, and AI-driven applications for defence and aerospace ecosystems.
- Received technical training in **Advanced Drone Technology** (Air Taxi & Defence UAVs), Aircraft Design, Rocketry Design, EV Technology & Charging Infrastructure, and Robotics Engineering.

Abhyudaya, S. P. Jain Institute of Management and Research

April 2025 – Feb. 2026

Volunteer

Mumbai, Maharashtra

- Mentored and taught underprivileged students evaluated examination papers and assisted in organizing and conducting educational workshops tailored to different learning levels and academic needs.

Projects

Indoor AR Navigation System | *React Native, Node.js, Express, Firebase, ViroReact*

Nov. 2025

- Built a full-stack indoor navigation app with real-time AR guidance and 3D directional overlays, integrated SLAM tracking and QR-based positioning for accurate AR navigation and dynamic route management
- Implemented the A* pathfinding algorithm on a Node.js/Express backend to compute optimal routes using graph-based waypoints.

Medical Care Management System | *React, TypeScript, Vite, Tailwind CSS, TanStack Query*

July 2025

- Developed a healthcare management platform with role-based access for Patients, Doctors, Receptionists, and Administrators.
- Built automated patient queue management with priority scheduling, token generation, and wait-time estimation. Integrated TanStack Query for efficient data synchronization

Certifications & Research

Paper accepted at IEEE AISIIS 2026: *Poisoning-Resistant Federated Intrusion Detection System.*

Completed the **Advanced Aircraft Control Systems with MATLAB/Simulink** certification from **IIT Kanpur**.

Certified in **Remote Sensing and GIS** by **IIRS, ISRO**.

Certified in **OCI for Data Science** by **Oracle**.

Certifications & Achievements

Ranked **5th nationally** at **India Innovates**, a National Hackathon organized by the **Municipal Corporation of Delhi**.

Certified A2 German Speaker, Goethe-Institut; currently pursuing **B1-level** certification.

Head of Public Relations, **Astrophysics Club**, Sardar Patel Institute of Technology.